

실습 가이드: OpenFOAM Laminar reacting flow

2026 한국연소학회 가을 Tutorial
Next-Gen Combustion

Turbulent reacting flow

● Governing equations of multiComponentFluid

- A solver for transient, compressible, **turbulent** reacting flow included in the standard OpenFOAM framework.
- Here, the **unity Schmidt number** (Sc) and **unity Lewis number** (Le) assumptions are applied for simplification.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0,$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \nabla \cdot \boldsymbol{\tau} = -\nabla p,$$

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (\mu_{\text{eff}} \nabla Y_i) + w_i$$

$$\frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{u} h) + \frac{\partial(\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{u} K) = \frac{\partial p}{\partial t} + \nabla \cdot (\alpha_{\text{eff}} \nabla h) + \sum h_i^0 w_i$$

$$Sc = \frac{\nu}{D} = \frac{\text{kinematic viscosity}}{\text{mass diffusivity}}$$

$$Le = \frac{\alpha}{D} = \frac{\text{thermal diffusivity}}{\text{mass diffusivity}}$$

Laminar reacting flow

- **Governing equations of DTLreactingFoam solver [*]**

- A solver for transient, compressible, **laminar** reacting flow included in the DTLreactingFoam framework.
- Designed to simulate reacting flows with detailed **chemistry**, **heat release**, and **turbulence** interactions.

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0,$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) - \nabla \cdot \boldsymbol{\tau} = -\nabla p,$$

$$\frac{\partial \rho Y_i}{\partial t} + \nabla \cdot (\rho \mathbf{u} Y_i) = \nabla \cdot (\rho \mathbf{V}_i \nabla Y_i) + w_i$$

$$\begin{aligned} & \frac{\partial(\rho h)}{\partial t} + \nabla \cdot (\rho \mathbf{u} h) + \frac{\partial(\rho K)}{\partial t} + \nabla \cdot (\rho \mathbf{u} K) - \frac{\partial p}{\partial t} \\ &= \nabla \cdot \left(\frac{\lambda}{c_p} \nabla h \right) - \nabla \cdot \left(\rho \sum_{i=1}^N h_i Y_i \mathbf{V}_i \right) + \sum h_i^0 w_i \end{aligned}$$

- Diffusion velocity

$$\mathbf{V}_i = -\frac{D_{i,\text{mix}}}{X_i} \nabla X_i$$

Laminar reacting flow

● Detailed Transport Model (DTM) [*]

- Transport properties such as viscosity μ , thermal conductivity λ , and binary diffusivity D_{ji} are evaluated based on **standard kinetic theory** [*]

❖ Viscosity

$$\mu_k = \frac{5}{16} \frac{\sqrt{\pi m_k k_B T}}{\pi \sigma_k^2 \Omega^{(2,2)*}};$$

❖ Thermal conductivity

$$\lambda_k = \frac{\mu_k}{m_k} (f_{\text{trans}} C_{v,\text{trans}} + f_{\text{rot}} C_{v,\text{rot}} + f_{\text{vib}} C_{v,\text{vib}})$$

❖ Mass diffusivity

$$D_{i,\text{mix}} = \frac{1 - Y_i}{\sum_{j \neq i}^K X_j / D_{ji}}; \quad \text{where} \quad D_{ji} = \frac{3}{16} \frac{\left(\frac{2 \pi k_B^3 T^3}{m_{ji}} \right)^{1/2}}{p \pi \sigma_{ji}^2 \Omega^{(1,1)*}}$$

[*] Kee et al., SAND96-8216, Sandia National Laboratories (1996).

Laminar reacting flow

- **Detailed Transport Model (DTM) [*]**

- Mixture properties are obtained based on a mixture-averaged model [*]

- ❖ Viscosity

$$\mu = \sum_{k=1}^K \frac{X_k \mu_k}{\sum_{j=1}^K X_j \Phi_{kj}}$$

- ❖ Thermal conductivity

$$\lambda = \frac{1}{2} \left(\sum_{k=1}^K X_k \lambda_k + \frac{1}{\sum_{k=1}^K X_k / \lambda_k} \right)$$

Where the coefficient

$$\Phi_{kj} = \frac{1}{\sqrt{8}} \left(1 + \frac{m_k}{m_j} \right)^{-\frac{1}{2}} \left(1 + \left(\frac{\mu_k}{\mu_j} \right)^{\frac{1}{2}} \left(\frac{m_k}{m_j} \right)^{\frac{1}{4}} \right)^2$$

Laminar reacting flow

● Existing framework for laminar reacting flow

- laminarSMOKE¹
- OF-Cantera:
 - reactingDNS²,
 - CTrreactingFoam³,
 - DeepFlame⁴
- reactingFoam-SCI⁵,
- combustionFoam⁶
- laminarReactingFoam⁷
- standardReactingFoam⁸

- [1] A. Cuoci, A. Frassoldati, T. Faravelli, E. Ranzi, Combust. Flame 160 (2013) 870-886.
- [2] S. Zhong, F. Zhang, Z. Peng, F. Bai, Q. Du, Fuel 234 (2018) 1044-1054.
- [3] D. Zhou, H. Zhang, S. Yang, Aerospace 9 (2022).
- [4] R. Mao, M. Lin, Y. Zhang, T. Zhang, Z.-Q. J. Xu, Z. X. Chen, Comput. Phys. Commun. 291 (2023) 108842.
- [5] Q. Yang, P. Zhao, H. Ge, Comput. Fluids 190 (2019) 114-127.
- [6] T. Li, J. Pan, F. Kong, B. Xu, X. Wang, Comput. Fluids 213 (2020) 104718.
- [7] A. Dasgupta, E. Gonzalez-Juez, D. C. Haworth, Comput. Phys. Commun. 237 (2019) 219-229.
- [8] F. Wang, L. An, T. L. Chan, Comput. Phys. Commun. 309 (2025) 109472

Laminar reacting flow

● Limitations of existing framework

Advantage

- Provide a detailed transport model (DTM)
- Integrates with OF as a third-party package

Disadvantage

- Does not fully **support all native** OF functionality
- The lack of a **user guide** and **C++ class diagrams** limits code extensibility and flexibility.

laminarSMOKE

Advantage

- Provide detailed chemistry and transport through Cantera
- Integrates Cantera-based thermochemical models with OF

Disadvantage

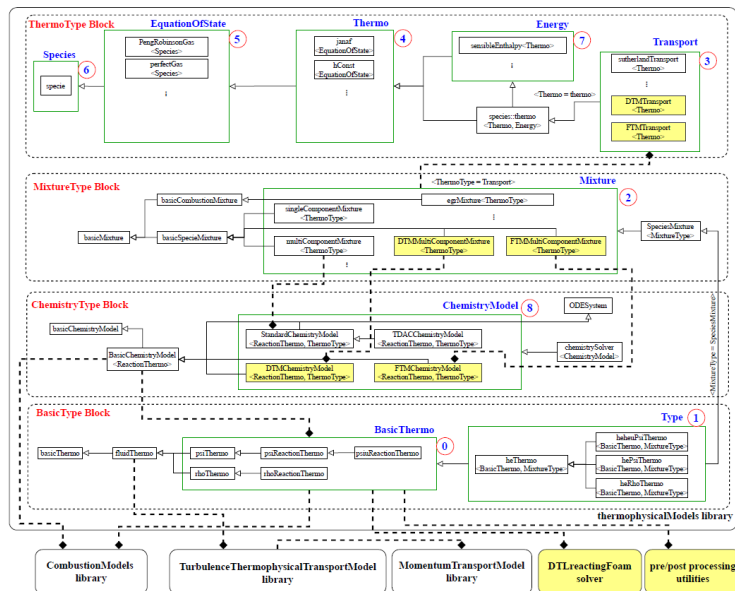
- **Dependency** on external library with **version compatibility** issues
- The **OF–Cantera interface** limits code extensibility and flexibility

Cantera coupled frameworks

Laminar reacting flow

- **DTLreactingFoam [*]:**

- Our group developed a **fully native** OF framework for DTM-based laminar reacting flow simulation while preserving **existing OF functionality**.
- Provides methods for reducing computational cost through **FTM** and **coTHERM** strategy.
- **Source code**, **User guide** and **tutorial cases** are available on GitHub repository **[**]**



DTLreactingFoam Public

 main

 1 Branch

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 Code

 danhnam11 Update README.md 2b1f551 · 3 weeks ago 4 Commits

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 README

DTLreactingFoam

latest release **v1.0.0**

contributions **welcome**

license **GPL-3.0**

DOI **10.1016/j.cpc.2026.110052**

Overview

A package for high fidelity simulations of laminar reacting flows in OpenFOAM with low computational cost, incorporating both the detailed transport model (DTM) and the polynomial fit transport model (FTM) based on the principle of kinetic gas theory [1]. To enhance computational efficiency, it was integrated with the time-correlated thermophysical property calculation (coTHERM) method. This technique can significantly reduce the computational cost of numerical simulations using DTM/FTM in OpenFOAM while preserving accuracy. Readers are referred to our paper for all validation data.

- Class diagram of DTLreactingFoam
- DTLreactingFoam GitHub repository ^[**]

[*] D. N. Nguyen, J. H. Lee, and C. S. Yoo, *Comput. Phys. Commun.* 322, 110052 (2026).

[**] <https://github.com/CCERL/DTLreactingFoam>

Laminar reacting flow

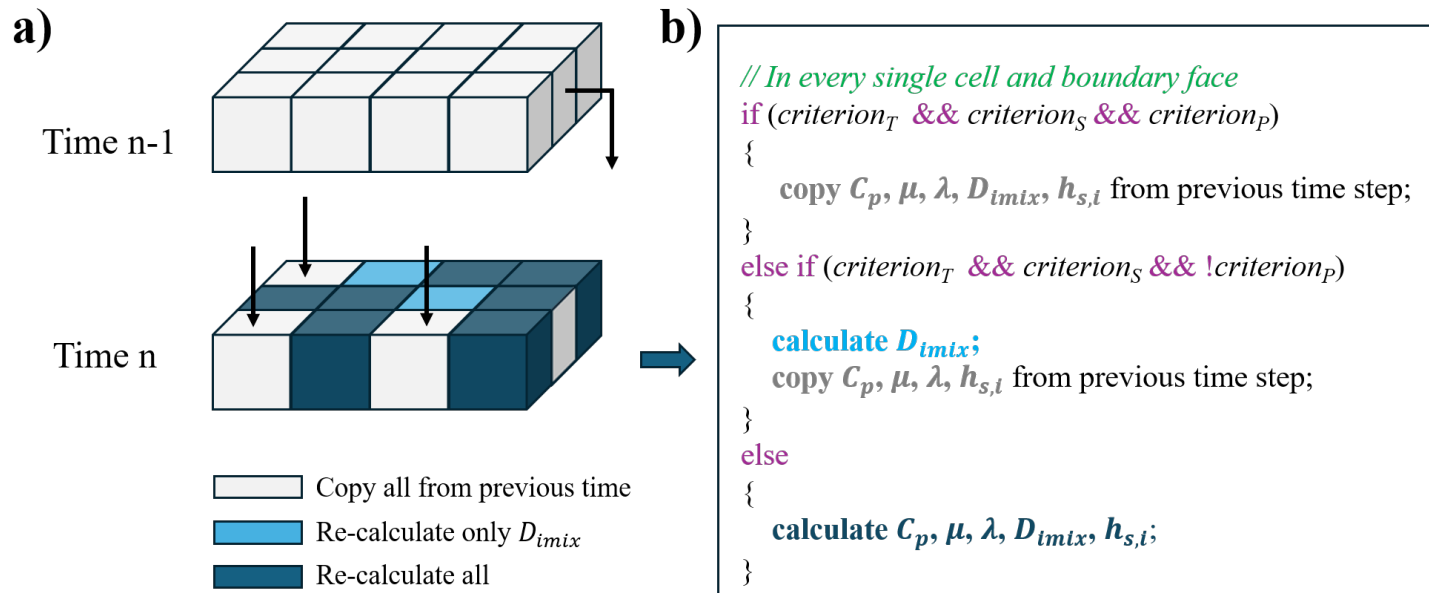
● Methods for reducing computational cost

1. Polynomial-fit transport models (FTM):

$$\ln \mu_k = \sum_{n=1}^N a_{n,k} (\ln T)^{n-1}; \quad \ln \lambda_k = \sum_{n=1}^N b_{n,k} (\ln T)^{n-1}; \quad \ln D_{jk} = \sum_{n=1}^N d_{n,jk} (\ln T)^{n-1}$$

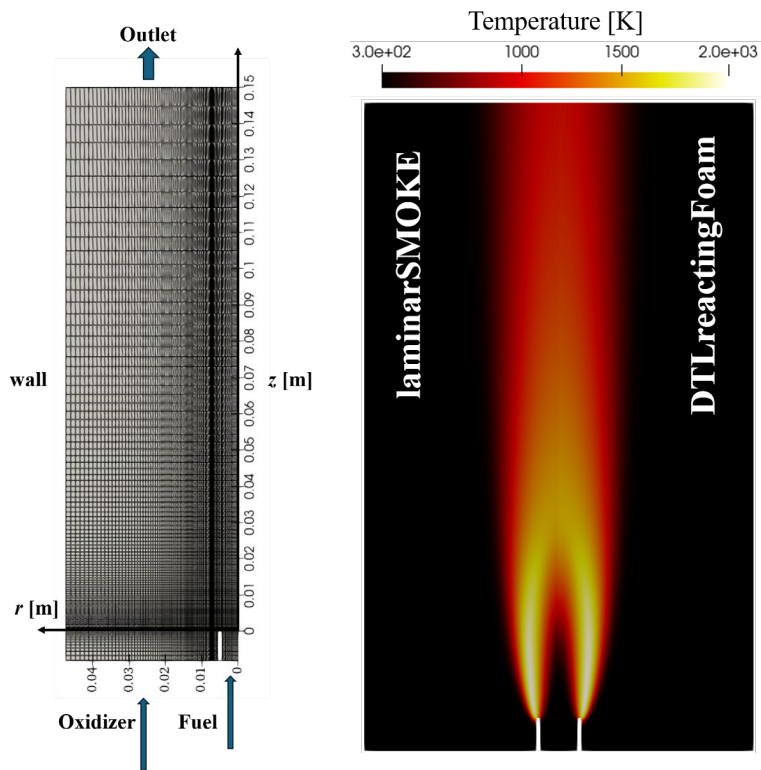
Using third-order polynomial fits: $N = 4$

2. Time correlation for thermophysical properties (**coTHERM**):



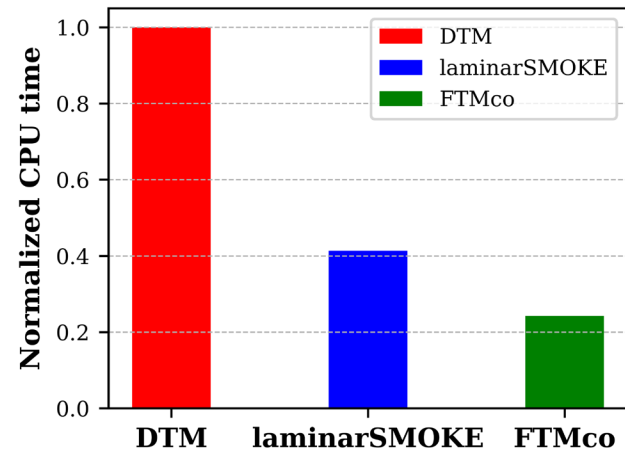
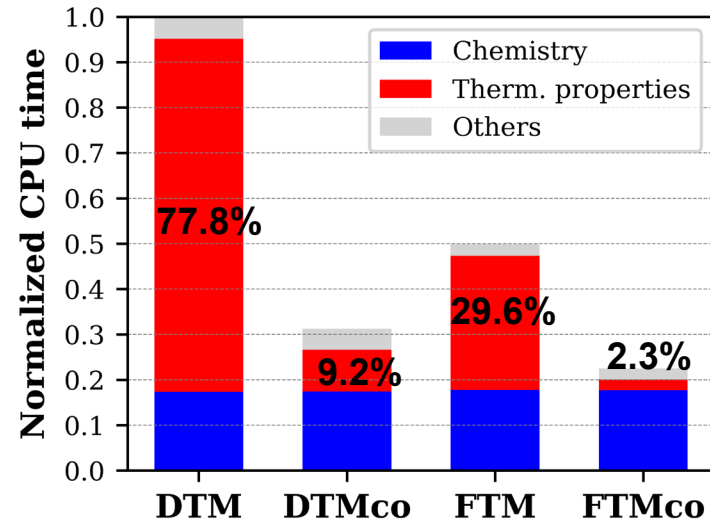
Laminar reacting flow

● Computational efficiency of DTLreactingFoam



Threshold values

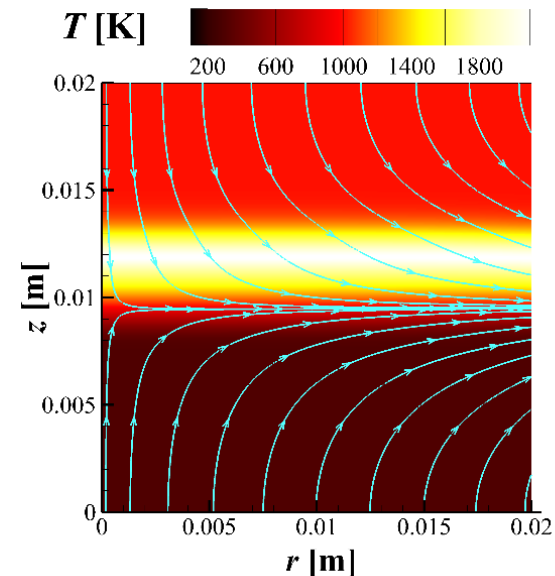
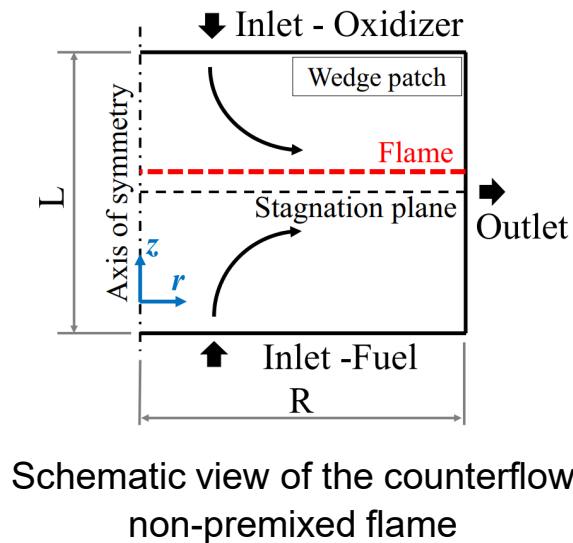
ε_T [K]	ε_S [-]	ε_p [Pa]
0.2	1E-3	100



Laminar reacting flow

● Tutorial 3: Counter flow flame

- 2D **laminar** non-premixed counter flow flame
- Fuel: 50% H_2 + 50% N_2 , Oxidizer: 21% O_2 + 79% N_2
- Inlet velocity U_{fuel} : 0.5 m/s, U_{oxi} : 0.5 m/s
- Pressure: 1atm



The temperature isocontours of 2-D laminar counterflow non-premixed flame

Laminar reacting flow

- Preprocessing

- Get the tutorial files

Command Line

```
cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM
```

```
ghkrk@20225263-p01:~$ cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ ll
total 32
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 10:56 ./
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 10:56 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 0/
-rwxr-xr-x 1 ghkrk ghkrk  310 Oct  6 10:56 Allclean*
-rwxr-xr-x 1 ghkrk ghkrk  684 Oct  6 10:56 Allrun.DTM*
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 chemkin/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 constant/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 system/
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ |
```

Laminar reacting flow

● Preprocessing: Kinetic mechanism

- Detailed hydrogen oxidation mechanism of Li et al. [*].

Table I Detailed H₂/O₂ Reaction Mechanism

		<i>A</i>	<i>n</i>	<i>E</i>	Reference
H ₂ /O ₂ chain reactions					
1.	H + O ₂ = O + OH	3.55×10^{15}	-0.41	16.6	[16]
2.	O + H ₂ = H + OH	5.08×10^4	2.67	6.29	[44]
3.	H ₂ + OH = H ₂ O + H	2.16×10^8	1.51	3.43	[45]
4.	O + H ₂ O = OH + OH	2.97×10^6	2.02	13.4	[46]
H ₂ /O ₂ dissociation/recombination reactions					
5.	H ₂ + M = H + H + M ^a	4.58×10^{19}	-1.40	104.38	[47]
	H ₂ + Ar = H + H + Ar	5.84×10^{18}	-1.10	104.38	[47]
	H ₂ + He = H + H + He	5.84×10^{18}	-1.10	104.38	See text
6.	O + O + M = O ₂ + M ^a	6.16×10^{15}	-0.50	0.00	[47]
	O + O + Ar = O ₂ + Ar	1.89×10^{13}	0.00	-1.79	[47]
	O + O + He = O ₂ + He	1.89×10^{13}	0.00	-1.79	See text
7.	O + H + M = OH + M ^a	4.71×10^{18}	-1.0	0.00	[47]
8.	H + OH + M = H ₂ O + M ^b	3.8×10^{22}	-2.00	0.00	See text
Formation and consumption of HO ₂					
9.	H + O ₂ + M = HO ₂ + M ^c	k_0 6.37×10^{20}	-1.72	0.52	[19]
	H + O ₂ + M = HO ₂ + M ^d	k_0 9.04×10^{19}	-1.50	0.49	[19]
		k_∞ 1.48×10^{12}	0.60	0.00	[48]
10.	HO ₂ + H = H ₂ + O ₂	1.66×10^{13}	0.00	0.82	[1]
11.	HO ₂ + H = OH + OH	7.08×10^{13}	0.00	0.30	[1]
12.	HO ₂ + O = OH + O ₂	3.25×10^{13}	0.00	0.00	[34]
13.	HO ₂ + OH = H ₂ O + O ₂	2.89×10^{13}	0.00	-0.50	[34]
Formation and consumption of H ₂ O ₂					
14.	HO ₂ + HO ₂ = H ₂ O ₂ + O ₂ ^e	4.20×10^{14}	0.00	11.98	[49]
	HO ₂ + HO ₂ = H ₂ O ₂ + O ₂	1.30×10^{11}	0.00	-1.63	
15.	H ₂ O ₂ + M = OH + OH + M ^f	k_0 1.20×10^{17}	0.00	45.5	[50]
		k_∞ 2.95×10^{14}	0.00	48.4	[51]
16.	H ₂ O ₂ + H = H ₂ O + OH	2.41×10^{13}	0.00	3.97	[47]
17.	H ₂ O ₂ + H = H ₂ + HO ₂	4.82×10^{13}	0.00	7.95	[47]
18.	H ₂ O ₂ + O = OH + HO ₂	9.55×10^6	2.00	3.97	[47]
19.	H ₂ O ₂ + OH = H ₂ O + HO ₂ ^e	1.00×10^{12}	0.00	0.00	[52]
	H ₂ O ₂ + OH = H ₂ O + HO ₂	5.8×10^{14}	0.00	9.56	

[*] J. Li, Z. Zhao, A. Kazakov, and F. L. Dryer, *Int. J. Chem. Kinet.* **36**, 566–575 (2004).

Laminar reacting flow

- **Preprocessing: Kinetic mechanism**

- Convert CHEMKIN-format chemistry, thermodynamic, and transport data into OpenFOAM-compatible DTM thermodynamic and reaction file

Command Line

```
DTMchemkinToFoam \  
  chemkin/chem.inp \  
  chemkin/therm.dat \  
  chemkin/tran.dat \  
  chemkin/transportSutherland \  
  constant/reactions \  
  constant/thermo.DTM
```

Laminar reacting flow

- **Preprocessing: Kinetic mechanism**

- Convert Chemkin-format files into OF-format reaction and and thermodynamic file

Command Line

```
vi constant/thermo.DTM
```

```
species          9 ( H2 O2 O OH H2O H HO2 H2O2 N2 );

OH
{
  specie
  {
    molWeight      17.00737;
  }
  thermodynamics
  {
    Tlow           200;
    Thigh          6000;
    Tcommon        1000;
    highCpCoeffs   ( 2.86472886 0.00105650448 -2.59082758e-07 3.05218674e-11 -1.33195876e-15 3683.62875 5.70164073 );
    lowCpCoeffs    ( 4.12530561 -0.00322544939 6.52764691e-06 -5.79853643e-09 2.06237379e-12 3346.30913 -0.69043296 );
  }
  transport
  {
    As             1.512e-06;
    Ts             120;
    linearity       1;
    epsilonOverKb   80;
    sigma          2.75;
    dipoleMoment    0;
    alpha          0;
    Zrot            0;
  }
  elements
  {
    O              1;
    H              1;
  }
}
```

DTM parameters

Laminar reacting flow

● Preprocessing: Physical models

1. Enable detailed transport and configure CoTHERM algorithm.
2. Set the CoTHERM threshold for T , p , and Y_i .
3. Select the multicomponent mixture and detailed transport models.
4. Include thermo.DTM.

Command Line

```
vi constant/physicalProperties
```

```
/*----- C++ -----*/
FoamFile
{
    format      ascii;
    class       dictionary;
    location     "constant";
    object       physicalProperties;
}
// *****

usingDetailedTransportModel true;
usingCoTHERM false;
usingPreProcessingFTM false;
majorSpeciesForCoTHERM (H2 O2 H2O);

epsilonT 0.2; // [K]
epsilonP 100; // [Pa]
epsilonS 0.001; // [-], in mass fraction

thermoType
{
    type hePsiThermo;
    mixture DTMMulticomponentMixture;
    transport DTMTransport;
    thermo janaf;
    energy sensibleEnthalpy;
    equationOfState perfectGas;
    specie specie;
}

defaultSpecie N2;

#include "thermo.DTM"

// *****
```

Laminar reacting flow

- Preprocessing: Mesh Generation

Command Line

blockMesh

```
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM$ blockMesh
/*-----*/
      //
      //      F ield      | OpenFOAM: The Open Source CFD Toolbox
      //      O peration  | Website:  https://openfoam.org
      //      A nd        | Version:   12
      //      M anipulation|
/*-----*/
Build   : 12-86e126a7bc4d
Exec    : blockMesh
Date    : Oct 06 2026
Time    : 11:01:29
Host    : "20225263-p01"
PID     : 73088
I/O     : uncollated
Case    : /home/ghkrk/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/3.CounterFlowFlame.DTM
nProcs  : 1
sigFpe  : Enabling floating point exception trapping (FOAM_SIGFPE).
fileModificationChecking : Monitoring run-time modified files using timeStampMaster (fileModificationSkew 10)
allowSystemOperations : Allowing user-supplied system call operations

// *****
Create time
```

Laminar reacting flow

- **Preprocessing: Domain decomposition**

- Parallel computing

In system/decomposeParDict file

```

|*-----* C++ -*-----*
|
|  F ield      | OpenFOAM: The Open Source CFD Toolbox
|  O peration | Website:  https://openfoam.org
|  A nd       | Version:  12
|  M anipulation |
|
|*-----*
FoamFile
{
    format      ascii;
    class       dictionary;
    note        "mesh decomposition control dictionary";
    object      decomposeParDict;
}
// *****

numberOfSubdomains 4;

method                simple;

simpleCoeffs
{
    n                  (1 4 1);
    delta              0.001;
}

// *****

```

Command Line

decomposePar

Laminar reacting flow

- Preprocessing: Simulation control

Command Line

```
cd system
```

```
vi controlDict
```

```
/*----- C++ -----*/
//
//   F i e l d
//   O p e r a t i o n
//   A n d
//   M a n i p u l a t i o n
//
//   OpenFOAM: The Open Source CFD Toolbox
//   Website:  https://openfoam.org
//   Version:  12
//
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "system";
    object      controlDict;
}
// *****

application    foamRun;
solver          DTLreactingFoam;
startFrom      startTime;
startTime      0;
stopAt         endTime;
endTime        0.1;
deltaT         1e-6;
writeControl    adjustableRunTime;
writeInterval   0.005;
purgeWrite     0;
writeFormat     ascii;
writePrecision  6;
writeCompression off;
timeFormat      general;
timePrecision   6;
runTimeModifiable true;
adjustTimeStep  yes;
```

Laminar reacting flow

- **Solving: Run the simulation**
 - Use DTLreactingFoam solver

Command Line

```
mpirun -np 4 foamRun -parallel
```

Laminar reacting flow

● Post-Processing

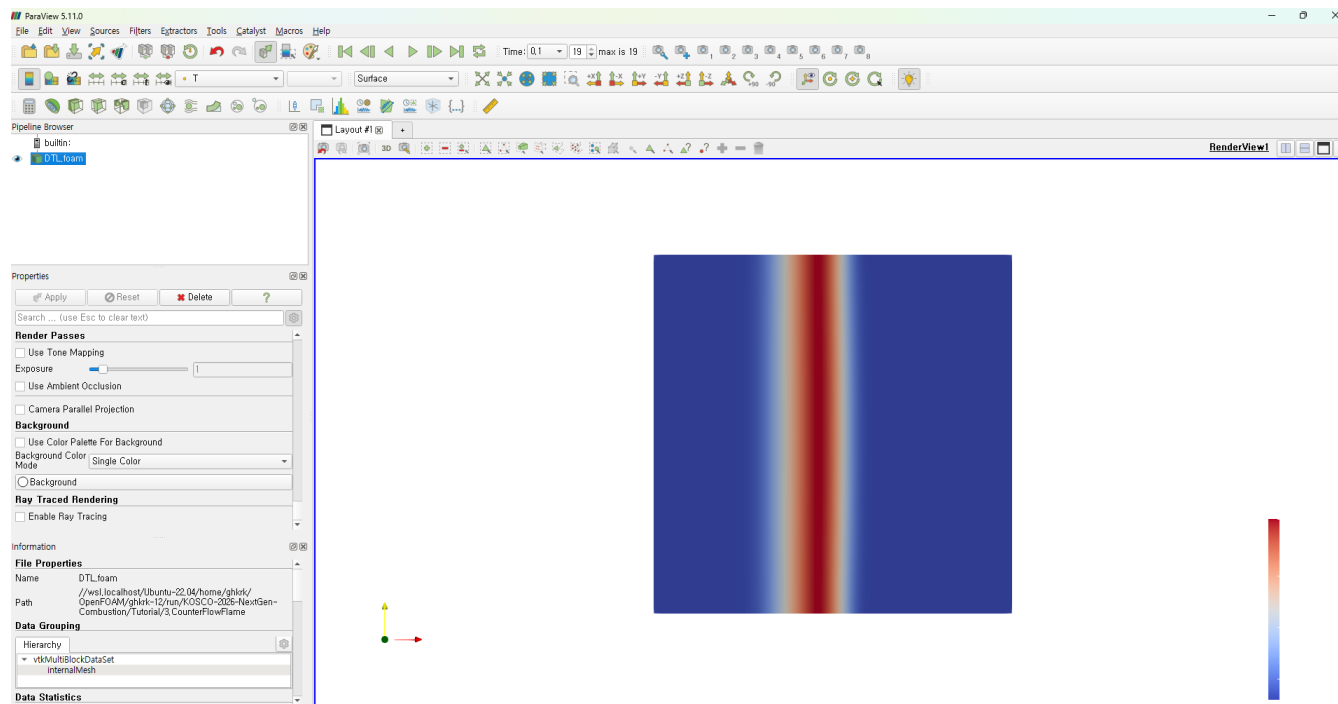
Command Line

```
reconstructPar
```

```
touch DTL.foam
```

→ Copy & paste files
to C: or D: drive

→ Open foam file
with paraview



Laminar reacting flow

- Preprocessing for FTM

- Get the tutorial files

Command Line

```
cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/4.CounterFlowFlame.FTM
```

```
ghkrk@20225263-p01:~$ cd $FOAM_RUN/KOSCO-2026-NextGen-Combustion/Tutorial/4.CounterFlowFlame.FTM
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/4.CounterFlowFlame.FTM$ ll
total 36
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 10:56 ./
drwxr-xr-x 6 ghkrk ghkrk 4096 Oct  6 10:56 ../
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 0/
-rwxr-xr-x 1 ghkrk ghkrk  328 Oct  6 10:56 Allclean*
-rwxr-xr-x 1 ghkrk ghkrk 1258 Oct  6 10:56 Allprepare.FTM*
-rwxr-xr-x 1 ghkrk ghkrk 1166 Oct  6 10:56 Allrun.FTM*
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 chemkin/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 constant/
drwxr-xr-x 2 ghkrk ghkrk 4096 Oct  6 10:56 system/
ghkrk@20225263-p01:~/OpenFOAM/ghkrk-12/run/KOSCO-2026-NextGen-Combustion/Tutorial/4.CounterFlowFlame.FTM$ |
```

Laminar reacting flow

- **Preprocessing for FTM**

- Enable FTM preprocessing.
- Set the transport model to `preprocessingFTMTransport`.

Command Line

```
vi constant/physicalProperties
```

```

/*-----*- C++ -*/
=====
\ \ \ \ \ F i e l d       | OpenFOAM: The Open Source CFD Toolbox
\ \ \ \ \ O p e r a t i o n   | Website: https://openfoam.org
\ \ \ \ \ A n d               | Version: 12
\ \ \ \ \ M a n i p u l a t i o n |
=====
/*-----*/

FoamFile
{
    format      ascii;
    class       dictionary;
    location     "constant";
    object       physicalProperties;
}
// ***** //

usingDetailedTransportModel true;
usingCoTHERM false;
usingPreProcessingFTM true;
majorSpeciesForCoTHERM (H2 O2 H2O);

epsilonT          0.2; // [K]
epsilonP          100; // [Pa]
epsilonS          0.001; // [-], in mass fraction

thermoType
{
    type            hePsiThermo;
    mixture         DTMMulticomponentMixture;
    transport       preprocessingFTMTransport
    thermo          janaf;
    energy          sensibleEnthalpy;
    equationOfState perfectGas;
    specie          specie;
}

defaultSpecie N2;

#include "thermo.DTM"

// ***** //

```

Laminar reacting flow

- Preprocessing for FTM

- Generate thermo.FTM containing the FTM transport coefficients.

Command Line

```
blockMesh
```

```
FTMchemkinToFoam
```

```
vi constant/thermo.FTM
```

```
OH
{
  transport
  {
    muCoeffs      (-17.3099 1.90275 -0.170195 0.00767127);
    kappaCoeffs   (4.12383 -3.89714 0.629982 -0.0280239);
    DijCoeffs     (
      (-20.112 2.28781 -0.0847962 0.00377877)
      (-23.4526 3.14265 -0.197273 0.00872065)
      (-22.376 2.88806 -0.164634 0.00732455)
      (-9.21034 -6.44736e-26 9.25847e-27 -4.40439e-28)
      (-28.6142 5.15624 -0.446304 0.0190235)
      (-22.5274 3.44613 -0.235979 0.010369)
      (-23.4579 3.14265 -0.197273 0.00872065)
      (-23.463 3.14265 -0.197273 0.00872065)
      (-23.2325 3.04984 -0.185302 0.00820502)
    );
  }
}
```

Laminar reacting flow

- Preprocessing for FTM

- Generate thermo.FTM containing the FTM transport coefficients.

Command Line

```
blockMesh
```

```
FTMchemkinToFoam
```

```
vi constant/thermo.FTM
```

```
OH
{
  transport
  {
    muCoeffs      (-17.3099 1.90275 -0.170195 0.00767127);
    kappaCoeffs   (4.12383 -3.89714 0.629982 -0.0280239);
    DijCoeffs     (
      (-20.112 2.28781 -0.0847962 0.00377877)
      (-23.4526 3.14265 -0.197273 0.00872065)
      (-22.376 2.88806 -0.164634 0.00732455)
      (-9.21034 -6.44736e-26 9.25847e-27 -4.40439e-28)
      (-28.6142 5.15624 -0.446304 0.0190235)
      (-22.5274 3.44613 -0.235979 0.010369)
      (-23.4579 3.14265 -0.197273 0.00872065)
      (-23.463 3.14265 -0.197273 0.00872065)
      (-23.2325 3.04984 -0.185302 0.00820502)
    );
  }
}
```

Laminar reacting flow

● Preprocessing for FTM

- Disable FTM preprocessing and enable CoTHERM.
- Select FTMMulticomponentMixture and FTMTransport as the mixture and transport models.
- Include thermo.FTM containing the generated transport coefficients.

Command Line

```
vi constant/physicalProperties
```

```
/*----- C++ -----*/
FoamFile
{
    format      ascii;
    class       dictionary;
    location    "constant";
    object      physicalProperties;
}
// *****

usingDetailedTransportModel true;
usingCoTHERM true;
usingPreProcessingFTM false;
majorSpeciesForCoTHERM (H2 O2 H2O);

epsilonT      0.2; // [K]
epsilonP      100; // [Pa]
epsilonS      0.001; // [-], in mass fraction

thermoType
{
    type        hePsiThermo;
    mixture     FTMMulticomponentMixture;
    transport    FTMTransport;
    thermo       janaf;
    energy       sensibleEnthalpy;
    equationOfState perfectGas;
    specie       specie;
}

defaultSpecie N2;

#include "thermo.FTM"

// *****
```

Laminar reacting flow

- **Solving: Run the simulation with FTM**
 - Use DTLreactingFoam solver

Command Line

```
decomposePar
```

```
mpirun -np 4 foamRun -parallel
```